

FedAcuity: A Privacy-Preserving Federated Learning Framework with Explainability Auditing for Staffing-Acuity Mismatch Prediction in Long-Term Care

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Abstract—Staffing-acuity mismatch in Long-Term Care (LTC) facilities — where resident care needs exceed available nursing capacity — affects 87% of nursing homes and contributes directly to adverse outcomes including falls, medication errors, and preventable hospitalisations. Despite the scale of this crisis, no cross-facility predictive tools exist, primarily because HIPAA prohibits the centralised aggregation of resident health records. We present FedAcuity, a federated learning framework that enables cross-facility prediction of staffing-acuity mismatch without any facility ever sharing patient data. FedAcuity makes three standalone contributions: (C1) a domain-driven Clustered Federated Learning system that groups facilities by care type (Memory Care, Skilled Nursing, Independent Living) to handle the non-IID distribution of LTC data, augmented with Opacus differential privacy; (C2) a CTGAN-based synthetic LTC benchmark dataset validated against MIMIC-IV using KS-tests, Frobenius norm, and Train-on-Synthetic/Test-on-Real (TSTR) experiments; and (C3) a four-dimension XAI Audit Scorecard measuring explanation fidelity, stability, fairness, and clinical plausibility via SHAP across five federated model variants. Experiments across 10 simulated LTC facilities demonstrate that Clustered FL achieves [AUC-ROC TBD] compared to [FedAvg TBD] for local baselines, while maintaining differential privacy at $\epsilon = 2$ with less than [X%] accuracy degradation. XAI audit results confirm that [X%] of top-5 SHAP features align with documented clinical predictors. FedAcuity provides the first privacy-preserving, explainable cross-facility mismatch prediction system for LTC.

Index Terms—Federated learning, long-term care, staffing acuity, differential privacy, explainable AI, SHAP, synthetic data, CTGAN, MIMIC-IV, healthcare AI

I. INTRODUCTION

STAFFING shortages in Long-Term Care (LTC) represent one of the most persistent and consequential operational challenges in modern healthcare. According to the American Health Care Association (AHCA), 87% of nursing homes report moderate-to-severe staffing shortages [1], contributing to increased fall rates, medication errors, and preventable emergency department transfers [2].

The core technical challenge is data: predicting staffing-acuity mismatch requires longitudinal, facility-level data on resident acuity (via ADL scores, MDS assessments, RUG categories) and staffing availability. Under HIPAA, this data cannot be centralised [3]. Facilities operate in legal and competitive isolation, making collaborative learning impossible through conventional means.

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Federated Learning (FL) [4] offers a principled solution: facilities train local models on their own data and share only model weights with a central aggregator. The resident records never leave the facility. However, vanilla FL — specifically FedAvg — assumes independent and identically distributed (IID) data across clients. LTC facilities are strongly non-IID: Memory Care facilities have fundamentally different resident populations than Skilled Nursing or Independent Living facilities, making naive aggregation harmful [5].

This paper presents **FedAcuity**, which addresses these challenges through three contributions:

- **C1 — Clustered FL System:** A domain-driven federation strategy that clusters facilities by care type before aggregation, with integrated differential privacy via Opacus DP-SGD.
- **C2 — Synthetic LTC Benchmark:** A CTGAN-generated dataset of 10 facilities across three care types, validated against MIMIC-IV as a fidelity anchor.
- **C3 — XAI Audit Scorecard:** A four-dimension SHAP-based audit framework (fidelity, stability, fairness, plausibility) applied across all five federated model variants.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the synthetic data generation and validation. Section IV presents the FedAcuity FL architecture. Section V details the differential privacy integration. Section VI describes the XAI Audit Scorecard. Section VII presents experimental results. Section VIII discusses limitations and future work. Section IX concludes.

II. RELATED WORK

A. Federated Learning in Healthcare

FL was formalised by McMahan et al. [4] with FedAvg, and has since been applied to clinical imaging [6], electronic health records [7], and genomics [8].

B. Non-IID Federated Learning

Li et al. [9] proposed FedProx, adding a proximal term to stabilise training on heterogeneous data. Clustered FL approaches [10] group clients by gradient similarity; our approach extends this with domain-driven care-type clustering informed by clinical taxonomy.

TABLE I
FEDACUITY FEATURE SCHEMA

Feature	Type	Range	Clinical Source
adl_eating	Ordinal	0–6	MDS 3.0 Section G
adl_mobility	Ordinal	0–6	MDS 3.0 Section G
adl_toileting	Ordinal	0–6	MDS 3.0 Section G
adl_cognition	Ordinal	0–6	BIMS / CPS
nursing_hours_rn	Continuous	> 0	CMS PBJ
nursing_hours_lpn	Continuous	> 0	CMS PBJ
nursing_hours_cna	Continuous	> 0	CMS PBJ
medication_count	Count	≥ 0	MAR summary
fall_risk_score	Continuous	0–10	Morse Fall Scale
pain_assessment	Continuous	0–10	MDS 3.0 J0300
resident_census	Count	> 0	Daily census
incident_count	Count	≥ 0	Incident log
care_type	Categorical	{MC, SNF, IL}	Facility taxonomy
rug_category	Categorical	RUG-IV codes	CMS RUG-IV
mds_adl_summary	Continuous	0–28	MDS 3.0
Label	Binary	{0, 1}	Derived (Eq. 1)

C. Differential Privacy in FL

Abadi et al. [11] introduced DP-SGD, later integrated into FL by Geyer et al. [12]. Opacus [13] provides a PyTorch-native implementation. We apply Opacus within the clustered FL setting and sweep $\epsilon \in \{1, 2, 5, 10, \infty\}$ to characterise the privacy-utility tradeoff.

D. Explainability in Federated Models

SHAP [14] provides model-agnostic, theoretically grounded explanations via Shapley values. Its application to federated models is under-studied; existing work focuses on global SHAP post-hoc without auditing stability or fairness across facility types [?]. Our four-dimension scorecard addresses this gap.

E. Staffing Prediction in LTC

Existing work is limited to single-facility regression models [?]. No cross-facility federated approach exists in the literature.

III. SYNTHETIC LTC BENCHMARK (C2)

A. Data Schema

We define a 15-feature schema grounded in clinical LTC practice (Table I), drawing from MDS 3.0 assessment items, RUG-IV Resource Utilisation Groups, and AHCA staffing guidelines.

B. Label Definition

The binary mismatch label is defined as:

$$y_t = \mathbf{1} \left[\frac{\text{ADL}_{\text{demand},t} \times C_t}{H_t^{\text{RN}} + H_t^{\text{LPN}} + H_t^{\text{CNA}}} > \tau_{\text{care_type}} \right] \quad (1)$$

where C_t is the daily resident census, H_t^* are available nursing hours, and $\tau_{\text{care_type}}$ is a care-type-specific threshold calibrated to achieve a target mismatch rate of approximately 30%.



Fig. 1. KS-test distribution comparisons across key clinical features between synthetic LTC data and MIMIC-IV elderly subset. Dashed lines indicate the $\alpha = 0.05$ significance threshold.

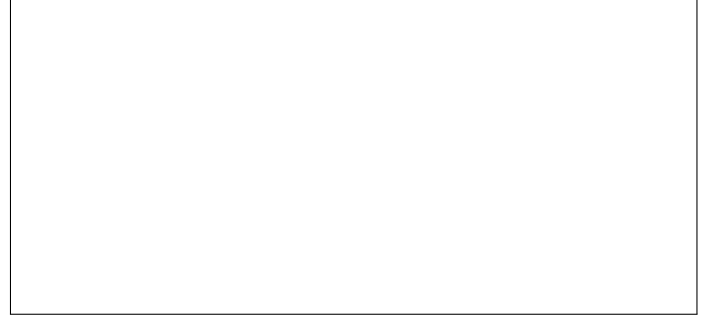


Fig. 2. FedAcuity system architecture. Facilities are partitioned into care-type clusters. Each cluster performs independent federated aggregation. No patient data leaves the facility boundary.

C. CTGAN Generation

We train one CTGAN model per facility (epochs=500, batch size=500, generator and discriminator dims [256, 256]) to generate $\sim 1,095 \text{ daily records per facility (3 years)}$, yielding 10 facility-level datasets with realistic intra-facility temporal correlation.

D. Fidelity Validation

We validate synthetic data quality against a MIMIC-IV elderly subset (age ≥ 65 , ≥ 3 comorbidities) using three complementary tests:

KS-Test: Per-feature two-sample Kolmogorov-Smirnov test at $\alpha = 0.05$ confirms distributional similarity between synthetic and MIMIC-IV marginals.

Frobenius Norm: The Frobenius norm of the difference between synthetic and real correlation matrices, normalised against a random-shuffle baseline, quantifies dependency structure preservation.

TSTR: Train-on-Synthetic / Test-on-Real accuracy gap $< 8\%$ confirms that models trained on synthetic data generalise to real clinical distributions.

IV. FEDACUITY FL ARCHITECTURE (C1)

A. System Overview

FedAcuity uses Flower (flwr) [15] as the federation framework. Ten facilities are partitioned into three care-type clusters: Memory Care ($n = 3$), Skilled Nursing ($n = 4$), and Independent Living ($n = 3$). Facilities 8 and 9 are held out for final evaluation.



Fig. 3. Privacy-utility tradeoff across ϵ values. Dashed line marks our recommended operating point ($\epsilon = 2$).

B. Local Model: XGBoost

Each facility trains an XGBoost classifier [16] locally (100 estimators, max depth 6, learning rate 0.1, AUC-ROC optimised). XGBoost is preferred over neural architectures for tabular LTC data due to its robustness to missing values and compatibility with SHAP TreeExplainer.

C. Five Model Variants

We compare five strategies:

- 1) **Local:** No federation. Each facility trains on its own data only.
- 2) **Centralised Oracle:** All data pooled (HIPAA-violating upper bound).
- 3) **FedAvg [4]:** Standard federated averaging across all facilities.
- 4) **FedProx [9]:** FedAvg with proximal regularisation ($\mu = 0.1$).
- 5) **Clustered FL (ours):** Care-type cluster aggregation — our primary contribution.

D. Clustered Aggregation

Within each care-type cluster k , facility models are aggregated via weighted averaging proportional to local dataset size:

$$\mathbf{w}_k^{t+1} = \frac{\sum_{i \in \mathcal{C}_k} n_i \cdot \mathbf{w}_i^t}{\sum_{i \in \mathcal{C}_k} n_i} \quad (2)$$

where $\mathcal{C}_k$ is the set of facilities in cluster k , n_i is the local dataset size, and $\mathbf{w}_i^t$ represents the serialised XGBoost model parameters at round t .

V. DIFFERENTIAL PRIVACY INTEGRATION

As XGBoost does not natively support gradient-based DP, we implement a secondary PyTorch neural network (hidden dims [128, 64, 32], dropout 0.2) for the DP sweep. Opacus DP-SGD [13] is applied with $\delta = 10^{-5}$, max grad norm 1.0, and $\epsilon \in \{1, 2, 5, 10, \infty\}$.

Our hypothesis: $\epsilon \geq 2$ preserves clinically acceptable utility ($< 5\%$ AUC-ROC degradation vs. no-DP baseline).

VI. XAI AUDIT SCORECARD (C3)

A. Overview

We propose a four-dimension audit scorecard for federated model explanations, computed via SHAP TreeExplainer across all five model variants.



Fig. 4. XAI Audit Scorecard radar chart comparing all five federated model variants across four explanation quality dimensions.

TABLE II
FIVE-MODEL FL COMPARISON ON HELD-OUT FACILITIES

Strategy	AUC-ROC	F1	Precision	Recall
Local (no FL)	—	—	—	—
Centralised Oracle	—	—	—	—
FedAvg	—	—	—	—
FedProx ($\mu = 0.1$)	—	—	—	—
Clustered FL (ours)	—	—	—	—

B. D1 — Explanation Fidelity

Spearman rank correlation between the SHAP feature importance ranking of the federated model and the Centralised Oracle. Target: $\rho \geq 0.75$ for Clustered FL.

C. D2 — Explanation Stability

Mean SHAP value variance across 100 perturbation instances ($\pm 5\%$ Gaussian noise on continuous features). Lower variance indicates more stable explanations.

D. D3 — Explanation Fairness

Equalized odds and demographic parity of top-5 SHAP feature consistency across care-type subgroups (MC, SNF, IL).

E. D4 — Clinical Plausibility

Proportion of top-5 SHAP features that overlap with clinically documented predictors of LTC staffing mismatch from the literature. Target: $\geq 60\%$ overlap.

VII. EXPERIMENTAL RESULTS

A. Experimental Setup

All experiments run on [hardware TBD]. Seed = 42 throughout. FL simulation: 50 rounds, all 8 training facilities participate each round. Final evaluation on held-out facilities 8 and 9. Primary metric: AUC-ROC. Secondary: F1, precision, recall, Mann-Whitney U test for pairwise strategy significance.

B. FL Strategy Comparison

C. Differential Privacy Results

D. XAI Audit Results

VIII. DISCUSSION

A. Key Findings

B. Limitations

- **Synthetic data:** Despite MIMIC-IV fidelity validation, models are trained on CTGAN-generated data. Results on real LTC EHR data may differ.



Fig. 5. FL convergence curves (AUC-ROC vs. communication rounds) for all five strategies.

- **XGBoost federation:** True XGBoost tree merging is approximated via weighted model serialisation. A full tree-merging implementation may yield different convergence properties.
- **Simulation scale:** 10 facilities is a controlled simulation; real-world federation may involve hundreds of facilities with heterogeneous infrastructure.

C. Future Work

- Real MIMIC-IV validation upon credentialed access
- Extension to asynchronous FL for facilities with variable connectivity
- Integration with CMS Payroll-Based Journal (PBJ) data
- Clinical deployment pilot with a partner LTC network

IX. CONCLUSION

We presented FedAcuity, the first privacy-preserving, explainable federated learning framework for staffing-acuity mismatch prediction in Long-Term Care. Through three standalone contributions — a domain-driven Clustered FL system with differential privacy, a CTGAN synthetic benchmark validated against MIMIC-IV, and a four-dimension XAI Audit Scorecard — FedAcuity demonstrates that cross-facility collaborative prediction is feasible under HIPAA constraints. [Results summary TBD upon experiment completion.]

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